



Sungrae Hong 홍성래

June 2026



[Address] Daejeon, Republic of Korea

[Interested In] Medical AI

[Email] sun.hong@kaist.ac.kr

[Military Service]

• ROK III Corps 🇰🇷 / 20160816 - 20180515 (YYYYMMDD)

[Education]

• Ph. D. Candidate KAIST Graduate School of Data Science (2023 ~)

• M.S. KAIST Graduate School of Data Science (2023)

• B.S. **Seoul** Nat'l Univ of Science & **Tech.** Industrial Engineering (2021)

• Daejeon High School (2014)

[Outbound Link] [Google Scholar](#) [linkedin](#)

Publications

2026

[CVPR] Every Error has Its Magnitude: Asymmetric Mistake Severity Training for Multiclass Multiple Instance Learning [\[Arxiv\]](#) [\[Proceeding\]](#)

- Sungrae Hong^o, Jiwon Jeong, Jisu Shin, Donghee Han, Sol Lee, Kyungeun Kim, Mun Yong Yi[†]

[CVPRW] DRCoD: Toward Robust Continual Learning of Diffusion Models for Tire Manufacturing Prototyping [\[Proceeding\]](#)

- Jisu Shin, Sol Lee, Sungrae Hong^o, A Young Kim, Youngbin You, Jeongheon Park, Jungsoo Oh, Mun Yong Yi[†]

- AI4RWC: The 2nd International Workshop on Vision Intelligence for Real-world Challenges

[SIGIR] Every Preference Has Its Strength: Injecting Ordinal Semantics into LLM-Based Recommenders [\[Arxiv\]](#)

- Jiwon Jeong^o, Donghee Han, Sungrae Hong, Woosung Kang, Mun Yong Yi[†]

[MICCAI] Missing Modality-Aware Calibration for Trustworthy Brain Tumor Segmentation [\[TBA\]](#)

- Sol Lee, Hyunji Kim, Sungrae Hong, Donghee Han, Mun Yong Yi[†]

- Early Accept (Top 9% among 4,601 Submissions)

[MICCAI] DiagMIL: Uncertainty-Gated Coarse-to-Fine Multi-Resolution MIL for WSI Diagnosis [\[TBA\]](#)

- Jiwon Jeong, Sungrae Hong, Donghee Han, Jisu Shin, Kyungeun Kim, Mun Yong Yi[†]

- Early Accept (Top 9% among 4,601 Submissions)

[MICCAI] RaLMPH: Reliability-aware Learning for Multi-Pathologist Harmonization in Whole-Slide Image Classification [\[Arxiv\]](#)

- Sungrae Hong, Jiwon Jeong, Soeun Cheon, Donghee Han, Sol Lee, Jisu Shin, Kyungeun Kim, Mun Yong Yi[†]

[MICCAI] Reasoning Trace Divergence: An Empirical Signal for Trustworthy Black-Box MLLMs in Histopathology Classification [TBA]

- Anastasiia Kazmina, Constantin Venhoff, Bryan Wong, Sungrae Hong, Yutong Xie, Mun Yong Yi[†]

[JIM (IF 7.4)]

Beyond Creative Generation: A Deep Generative Framework for High-Fidelity Pseudo-Virtual Prototyping and Design Evaluation [TBA]

- Sol Lee, Sungrae Hong, Jisu Shin, Mun Yong Yi[†]

- Journal of Intelligent Manufacturing

4 papers are under review.

2025

[WACV] Diagnose Like A REAL Pathologist: An Uncertainty-Focused Approach for Trustworthy Multi-Resolution Multiple Instance Learning [Arxiv] [Proceeding]

- Sungrae Hong[°], Sol Lee, Jisu Shin, Jiwon Jeong, Mun Yong Yi[†]

[MICCAIW/Oral] Priority-Aware Clinical Pathology Hierarchy Training for Multiple Instance Learning [Arxiv] [OpenReview]

- Sungrae Hong[°], Kyungeun Kim, Juhyeon Kim, Sol Lee, Jisu Shin, Chanjae Song, Mun Yong Yi[†]

- The 2nd MICCAI Student Board (MSB) EMERGE Workshop, September 23-27, 2025

- 🏆 **Best Paper Award**

[KCC] Calibrating MIL in Histopathology with Uncertainty-Aware Label Smoothing via Evidential Learning [Proceeding]

- Juheon Kim^{°*}, Sungrae Hong^{*}, Chanjae Song, Mun Yong Yi[†]

[KCC] Metadata-Aware Multimodal MIL with Reliable Calibration for Histopathology Diagnosis [Proceeding]

- Chanjae Song[°], Sungrae Hong, Junheon Kim, Minkyu Jeong, Mun Yong Yi[†]

- 🏆 **Best Presentation Award**

2024

[ISBI] Towards Classifying Histopathological Microscope Images as Time Series Data [Arxiv] [Proceeding]

- Sungrae Hong[°], Hyeongmin Park, Youngsin Ko, Sol Lee, Bryan Wong, Mun Yong Yi[†]

[ISBI] Rethinking Pre-Trained Feature Extractor Selection in Multiple Instance Learning for Whole Slide Image Classification [Proceeding]

- Bryan Wong, Sungrae Hong[°], Mun Yong Yi[†]

[AAAI Workshop] TireDiff: A Framework for Conditional Tire Footprint Images Generation from Manufacturing Tabular Data [Proceeding]

- Sol Lee[°], Jisu Shin, Sungrae Hong, Chanjae Song, Youngbin Yoo, Jeongheon Park, Jungsoo Oh, Mun Yong Yi[†]

- 4th Annual AAAI Workshop on AI to Accelerate Science and Engineering (AI2ASE), March 3-4, 2025

[WACV/Oral] Uncertainty-based Data-wise Label Smoothing for Calibrating Multiple Instance Learning in Histopathology Image Classification [Proceeding]

- Hyeongmin Park[◦], Sungrae Hong, Chanjae Song, Jongwoo Kim, Mun Yong Yi[†]

[KAIA] Sensitivity-Aware Fine-Tuning: Enhancing Parameter Importance with Hessian Connections [Proceeding]

- Nahyeon Lee[◦], Sol Lee, Sungrae Hong, Mun Yong Yi[†]

[KCC] Image Augmentation Based Self-supervised Learning for Music Genre Classification [Proceeding]

- Hyeongmin Park[◦], Sungrae Hong, Sol Lee, Taemi Kim, Mun Yong Yi[†]

2023

[Arxiv] CP-Dilatation: A Copy-and-Paste Augmentation Method for Preserving the Boundary Context Information of Histopathology Images [Arxiv]

- Sungrae Hong, Sol Lee, Mun Yong Yi[†]

2022

[KCC/Oral] MATE: The Multimodal Model Using Audio and Text for Emotion Recognition [Proceeding]

- Sungrae Hong[◦], Taemi Kim, Sol Lee, Jongwoo Kim, Mun Yong Yi[†]

- 🏆 **MSIT Minister's Award**

[KIIE] A Simple Cut Augmentation For Medical Pathology Image [Proceeding]

- Sungrae Hong[◦], Taemi Kim, Sol Lee, Mujin Kim, Bumho Lee, Youngsin Ko, Yumi Choi, Mun Yong Yi[†]

[KIIE/Oral] 약한 지도 학습 방식의 조직 병리학 이미지 세그멘테이션 [Proceeding]

- Taemi Kim[◦], Sungrae Hong, Sol Lee, Mujin Kim, Bumho Lee, Youngsin Ko, Yumi Choi, Mun Yong Yi[†]

[International Conference]

[International Journal]

[Domestic Conference]

†: Corresponding Author

◦: Speaker



Work Experience

Prof. Mun Yong Yi: Knowledge System Lab of ISE&GSDS, KAIST

Development of a Multimodal Artificial Intelligence-Based Computer-Aided Diagnosis System for Gastrointestinal Endoscopic Biopsies (Seegene Medical Foundation)

[Period] March 2024 - March 2026

[Experience]

- Project Manager
- Multiple Instance Learning using Multi-class and -Resolution WSIs
- Priority-Aware Mistake Severity Training for WSI
- Decision Making under Uncertainty

A Study on the Next Generation Medical Diagnosis System Based on AI (Seegene Medical Foundation)

[Period] July 2021 - February 2024

[Experience]

- Histopathology Whole Slide Segmentation
- Microscopy Pathology Image Classification under Weak Label and Noise
- Microscopy Diagnosis Team Manager

Prof. Sangheum Hwang: Data Intelligence Lab of IE, Seoultech

Development of Safety Monitoring Model Based on IoT for Subminiature Sensors (IITP)

[Period] July 2020 - July 2021

[Experience]

- Research Intern
- Time Series Modeling
- Anomaly Detection in Sequential Data



Awards

2025 MICCAI STUDENT BOARD (MSB) EMERGE WORKSHOP

- Best Paper Award
- Awarded NVIDIA[®] DGX Spark[™] as a Supplementary Prize [\[Link\]](#)
- Sungrae Hong[◦], Kyungeun Kim, Juhyeon Kim, Sol Lee, Jisu Shin, Chanjae Song, Mun Yong Yi[‡]

2024 KAIST ISE Research Day Competition

- Excellent Prize
- Sungrae Hong[◦], Hyeongmin Park, Mun Yong Yi[‡]

2023 Military AI Competition (MAICON) [\[github\]](#)

- Special Award by

2022 ETRI Human Understanding Artificial Intelligence Paper Competition [\[github\]](#)

- MSIT Minister's Award (과학기술정보통신부장관상)
- We also received **Best Prize(2nd)** from 2022 KAIST IE Research Day Competition
- Sungrae Hong[◦], Taemi Kim, Sol Lee, Jongwoo Kim, Mun Yong Yi[‡]

2020 Seoultech IE Capstone Competition

- Grand Prize
- Jaehyun Park[◦], Hyunjun Cha[◦], Sungrae Hong[◦], Gwangho GuK[‡]
(Same contribution, Name ascending in Korean)

◦: Speaker, ‡: Adviser



Reviewer

[2026]

CVPR 🏆 **Outstanding Reviewer**: Top 5% among 17,491 reviewers

MICCAI

- * * *
- * * *



Scholarship

[2022]

- 2022 KAIST IE Research Day Competition (Best Prize)

[2021]

- Academic Excellent Scholarship SeoulTech (GPA 4.5/4.5)

[2020]

- SeoulTech IE Capstone Competition (Grand Prize)
- Academic Excellent Scholarship SeoulTech (GPA 4.37/4.5)

[2019]

- Academic Scholarship SeoulTech (GPA 4.0/4.5)
- Academic Scholarship SeoulTech (GPA 4.11/4.5)
- Betterment in Ability Scholarship (Merit-Based)

[2015]

- Admission with Highest Distinction (Full Tuition)



Teaching Assistant

[2023]

- HanKook Tire AI Education, KAIST IE
 - DS545 Business Intelligence(BI), KAIST GSDS
-